

Research Statement

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Research Background

Large language models (LLMs) have rapidly become part of the daily lives of people and organizations, yet our understanding of how they work internally is still limited, and ensuring that they behave reliably and safely remains an open objective. My research pursues this goal from two complementary directions: designing defenses that keep LLMs safe under adversarial pressure, and understanding the internal mechanisms that make their safety hold or fail.

My interest in this objective grew out of my applied work with LLMs. My first paper, on multi-domain retrieval-augmented generation ([Rachidy et al., 2025](#)), addressed a practical trade-off between reliability and cost: a reranker fine-tuned on one specialized domain tends to degrade on others, while maintaining a separate reranker for each domain is costly to train and deploy. I introduced a Mixture-of-Experts reranker that routes each query to domain-specialized expert heads built on a shared encoder, achieving domain specialization within a single compact model. On agricultural question answering, it slightly outperformed commercial rerankers and an LLM-based reranker over a hundred times its size. This project gave me a practical foundation in fine-tuning and evaluating LLM systems. It also showed me how strongly a model’s output depends on its input, since irrelevant retrieved passages alone were enough to degrade the generator’s answers. If irrelevant context could do this by accident, an adversary could do it on purpose, and making models robust and safe became the focus of my subsequent work.

My main safety contribution is *Strategic Deflection* ([Rachidy et al., 2026](#)), a defense built on a change of objective. When refusal fails, the model should not comply but *deflect*, producing a response that appears compliant while steering toward a safe, related topic. This addresses a structural limitation of current safety training. Existing post-training defenses make refusal more persistent or more robust to adversarial manipulation, yet any defense whose last line is refusal remains exposed to an attacker who suppresses refusal itself. Logit manipulation attacks do exactly this, forcing an affirmative prefix and blocking refusal tokens during decoding. I trained deflection as a fallback using Contrastive Preference Optimization on response pairs that share the same forced prefix, so that the model learns to prefer a safe continuation precisely when refusal has already been bypassed. The defense substantially reduced attack success under logit manipulation, where refusal-based defenses remained largely ineffective, and held across model families and scales. It also generalized beyond the threat model it was designed for, to optimization-based, in-context, and obfuscation attacks as well as to an unseen language, while largely preserving general capabilities.

Looking inside the defended models, I found that token-level entropy follows a consistent pattern under attack. It is elevated during the forced prefix, indicating conflict with the imposed trajectory. Once the prefix ends, it spikes sharply, well above the levels of undefended models, which widens the set of plausible continuations at the point where the model redirects. It then returns to normal levels. I also found that standard safety classifiers misjudge safe deflections as harmful, often reacting to the compliant prefix rather than the content that follows, which exposes a blind spot in how safety is currently evaluated.

Research Vision

The question that drives my ongoing research is why LLM safety fails when it does. Defenses are usually designed and evaluated against the attacks known at the time, so a low attack success rate shows that a defense works against those attacks, not why it works or when it will break. Safety therefore tends to advance by patching vulnerabilities as they appear. I believe durable safety requires understanding such failures at their source, in the internal mechanisms that carry model behavior.

My long-term goal is to contribute to building AI systems that are both safe and transparent. As these systems become more capable and more autonomous, the most serious risks shift from harmful outputs to failures that are harder to observe: models that misrepresent their reasoning or intentions, pursue objectives other than those intended, or resist correction and oversight. Because these can hide behind acceptable outputs, I aim to address them through model internals, so that safety can be verified rather than only observed.

Selected References

Yassine Rachidy, Youssef Hmamouche, Faissal Sehbaoui, and Amal El Fallah Seghrouchni. Domain adaptive document reranking for retrieval augmented generation. In *2025 IEEE 37th International Conference on Tools with Artificial Intelligence (ICTAI)*, pages 1381–1387, 2025. doi: 10.1109/ICTAI66417.2025.00200.

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